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M/J Comprehensive Science 3 (#2002100)

Version: 2024 - And Beyond (current)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- [SC.8.E.5.1](#): Recognize that there are enormous distances between objects in space and apply our knowledge of light and space travel to understand this distance.
- [SC.8.E.5.2](#): Recognize that the universe contains many billions of galaxies and that each galaxy contains many billions of stars.
- [SC.8.E.5.3](#): Distinguish the hierarchical relationships between planets and other astronomical bodies relative to solar system, galaxy, and universe, including distance, size, and composition.
- [SC.8.E.5.4](#): Explore the Law of Universal Gravitation by explaining the role that gravity plays in the formation of planets, stars, and solar systems and in determining their motions.
- [SC.8.E.5.5](#): Describe and classify specific physical properties of stars: apparent magnitude (brightness), temperature (color), size, and luminosity (absolute brightness).
- [SC.8.E.5.6](#): Create models of solar properties including: rotation, structure of the Sun, convection, sunspots, solar flares, and prominences.
- [SC.8.E.5.7](#): Compare and contrast the properties of objects in the Solar System including the Sun, planets, and moons to those of Earth, such as gravitational force, distance from the Sun, speed, movement, temperature, and atmospheric conditions.
- [SC.8.E.5.8](#): Compare various historical models of the Solar System, including geocentric and heliocentric.

- [SC.8.E.5.9:](#) Explain the impact of objects in space on each other including:
 1. the Sun on the Earth including seasons and gravitational attraction
 2. the Moon on the Earth, including phases, tides, and eclipses, and the relative position of each body.
- [SC.8.E.5.10:](#) Assess how technology is essential to science for such purposes as access to outer space and other remote locations, sample collection, measurement, data collection and storage, computation, and communication of information.
- [SC.8.E.5.11:](#) Identify and compare characteristics of the electromagnetic spectrum such as wavelength, frequency, use, and hazards and recognize its application to an understanding of planetary images and satellite photographs.
- [SC.8.E.5.12:](#) Summarize the effects of space exploration on the economy and culture of Florida.
- [SC.8.L.18.1:](#) Describe and investigate the process of photosynthesis, such as the roles of light, carbon dioxide, water and chlorophyll; production of food; release of oxygen.
- [SC.8.L.18.2:](#) Describe and investigate how cellular respiration breaks down food to provide energy and releases carbon dioxide.
- [SC.8.L.18.3:](#) Construct a scientific model of the carbon cycle to show how matter and energy are continuously transferred within and between organisms and their physical environment.
- [SC.8.L.18.4:](#) Cite evidence that living systems follow the Laws of Conservation of Mass and Energy.
- [SC.8.N.1.1:](#) Define a problem from the eighth grade curriculum using appropriate reference materials to support scientific understanding, plan and carry out scientific investigations of various types, such as systematic observations or experiments, identify variables, collect and organize data, interpret data in charts, tables, and graphics, analyze information, make predictions, and defend conclusions.
- [SC.8.N.1.2:](#) Design and conduct a study using repeated trials and replication.
- [SC.8.N.1.3:](#) Use phrases such as "results support" or "fail to support" in science, understanding that science does not offer conclusive 'proof' of a knowledge claim.
- [SC.8.N.1.4:](#) Explain how hypotheses are valuable if they lead to further investigations, even if they turn out not to be supported by the data.

- [SC.8.N.1.5:](#) Analyze the methods used to develop a scientific explanation as seen in different fields of science.
- [SC.8.N.1.6:](#) Understand that scientific investigations involve the collection of relevant empirical evidence, the use of logical reasoning, and the application of imagination in devising hypotheses, predictions, explanations and models to make sense of the collected evidence.
- [SC.8.N.2.1:](#) Distinguish between scientific and pseudoscientific ideas.
- [SC.8.N.2.2:](#) Discuss what characterizes science and its methods.
- [SC.8.N.3.1:](#) Select models useful in relating the results of their own investigations.
- [SC.8.N.3.2:](#) Explain why theories may be modified but are rarely discarded.
- [SC.8.N.4.1:](#) Explain that science is one of the processes that can be used to inform decision making at the community, state, national, and international levels.
- [SC.8.N.4.2:](#) Explain how political, social, and economic concerns can affect science, and vice versa.
- [SC.8.P.8.1:](#) Explore the scientific theory of atoms (also known as atomic theory) by using models to explain the motion of particles in solids, liquids, and gases.
- [SC.8.P.8.2:](#) Differentiate between weight and mass recognizing that weight is the amount of gravitational pull on an object and is distinct from, though proportional to, mass.
- [SC.8.P.8.3:](#) Explore and describe the densities of various materials through measurement of their masses and volumes.
- [SC.8.P.8.4:](#) Classify and compare substances on the basis of characteristic physical properties that can be demonstrated or measured; for example, density, thermal or electrical conductivity, solubility, magnetic properties, melting and boiling points, and know that these properties are independent of the amount of the sample.
- [SC.8.P.8.5:](#) Recognize that there are a finite number of elements and that their atoms combine in a multitude of ways to produce compounds that make up all of the living and nonliving things that we encounter.
- [SC.8.P.8.6:](#) Recognize that elements are grouped in the periodic table according to similarities of their properties.
- [SC.8.P.8.7:](#) Explore the scientific theory of atoms (also known as atomic theory) by recognizing that atoms are the smallest unit of an

element and are composed of sub-atomic particles (electrons surrounding a nucleus containing protons and neutrons).

- SC.8.P.8.8: Identify basic examples of and compare and classify the properties of compounds, including acids, bases, and salts.
- SC.8.P.8.9: Distinguish among mixtures (including solutions) and pure substances.
- SC.8.P.9.1: Explore the Law of Conservation of Mass by demonstrating and concluding that mass is conserved when substances undergo physical and chemical changes.
- SC.8.P.9.2: Differentiate between physical changes and chemical changes.
- SC.8.P.9.3: Investigate and describe how temperature influences chemical changes.
- MA.K12.MTR.1.1: **Actively participate in effortful learning both individually and collectively.**

Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

- **MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.**

Mathematicians who demonstrate understanding by representing problems in multiple ways:

- Build understanding through modeling and using manipulatives.
- Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
- Progress from modeling problems with objects and drawings to using algorithms and equations.
- Express connections between concepts and representations.
- Choose a representation based on the given context or purpose.

Clarifications:

Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.
- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.

- **MA.K12.MTR.3.1: Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:

Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- **MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others.**

Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

- Communicate mathematical ideas, vocabulary and methods effectively.
- Analyze the mathematical thinking of others.
- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.
- Construct possible arguments based on evidence.

Clarifications:

Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

- **MA.K12.MTR.5.1: Use patterns and structure to help understand and connect mathematical concepts.**

Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

- Focus on relevant details within a problem.
- Create plans and procedures to logically order events, steps or ideas to solve problems.
- Decompose a complex problem into manageable parts.
- Relate previously learned concepts to new concepts.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.

Clarifications:

Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

- Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
- Support students to develop generalizations based on the similarities found among problems.
- Provide opportunities for students to create plans and procedures to solve problems.
- Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

• **MA.K12.MTR.6.1: Assess the reasonableness of solutions.**

Mathematicians who assess the reasonableness of solutions:

- Estimate to discover possible solutions.
- Use benchmark quantities to determine if a solution makes sense.
- Check calculations when solving problems.
- Verify possible solutions by explaining the methods used.
- Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to assess the reasonableness of solutions:

- Have students estimate or predict solutions prior to solving.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"

- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justifications.
- **MA.K12.MTR.7.1: Apply mathematics to real-world contexts.**

Mathematicians who apply mathematics to real-world contexts:

- Connect mathematical concepts to everyday experiences.
- Use models and methods to understand, represent and solve problems.
- Perform investigations to gather data or determine if a method is appropriate.
- Redesign models and methods to improve accuracy or efficiency.

Clarifications:

Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.
- Challenge students to question the accuracy of their models and methods.
- Support students as they validate conclusions by comparing them to the given situation.
- Indicate how various concepts can be applied to other disciplines.
- **ELA.K12.EE.1.1: Cite evidence to explain and justify reasoning.**

Clarifications:

K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing,

students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- [ELA.K12.EE.2.1](#): Read and comprehend grade-level complex texts proficiently.

Clarifications:

See [Text Complexity](#) for grade-level complexity bands and a text complexity rubric.

- [ELA.K12.EE.3.1](#): Make inferences to support comprehension.

Clarifications:

Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like “Why is the girl smiling?” or make predictions about what will happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- [ELA.K12.EE.4.1](#): Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

Clarifications:

In kindergarten, students learn to listen to one another respectfully.

In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think _____ because _____.” The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- [ELA.K12.EE.5.1](#): Use the accepted rules governing a specific format to create quality work.

Clarifications:

Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills

appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- [ELA.K12.EE.6.1](#): Use appropriate voice and tone when speaking or writing.

Clarifications:

In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd grade and beyond, students practice appropriate social and academic language to discuss texts.

- [ELD.K12.ELL.SC.1](#): English language learners communicate information, ideas and concepts necessary for academic success in the content area of Science.
- [ELD.K12.ELL.SI.1](#): English language learners communicate for social and instructional purposes within the school setting.

Course Information

General Notes

Laboratory investigations that include the use of scientific inquiry, research, measurement, problem solving, laboratory apparatus and technologies, experimental procedures, and safety procedures are an integral part of this course. The National Science Teachers Association (NSTA) recommends that at the middle school level, all students should have multiple opportunities every week to explore science laboratory investigations (labs). School laboratory investigations are defined by the National Research Council (NRC) as an experience in the laboratory, classroom, or the field that provides students with opportunities to interact directly with natural phenomena or with data collected by others using tools, materials, data collection techniques, and models (NRC, 2006, p. 3). Laboratory investigations in the middle school classroom should help all students develop a growing understanding of the complexity and ambiguity of empirical work, as well as the skills to calibrate and troubleshoot equipment used to make observations. Learners should understand measurement error; and have the skills to aggregate, interpret, and present the resulting data (NRC 2006, p. 77; NSTA, 2007).

Special Notes:

Instructional Practices